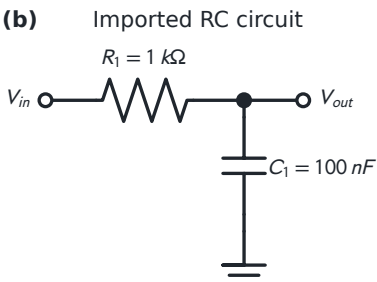
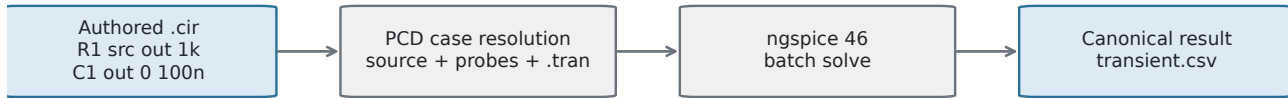


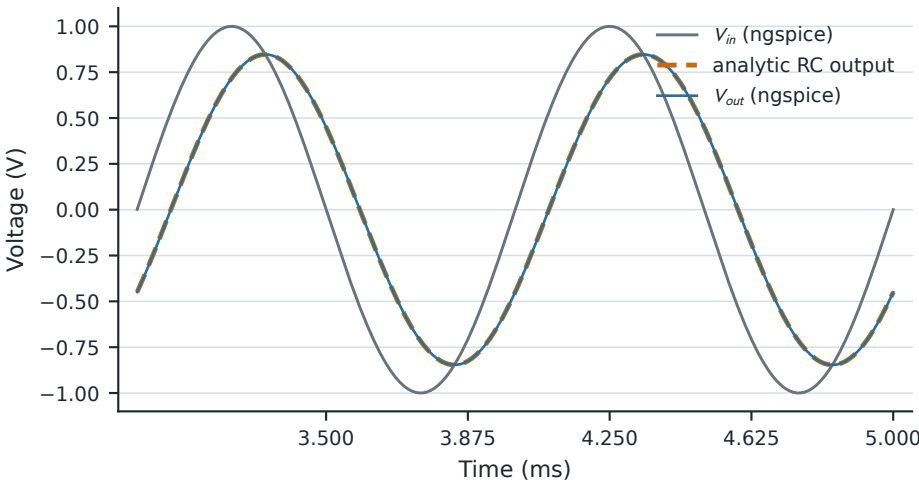
An authored netlist is resolved, solved, and returned as a canonical waveform

Direct 1 kHz RC low-pass example; output is compared with the independent steady-state transfer equation.

(a) Public calculation path



(c) Steady-state input and output

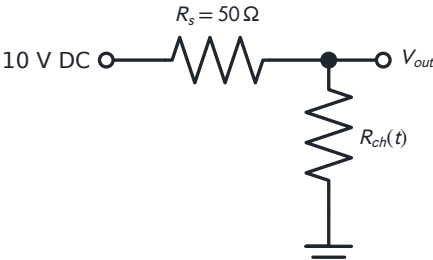


ngspice-46; last two cycles shown; analytic gain=0.846733, phase=-32.14 deg; max |error|=5.93 μ V.

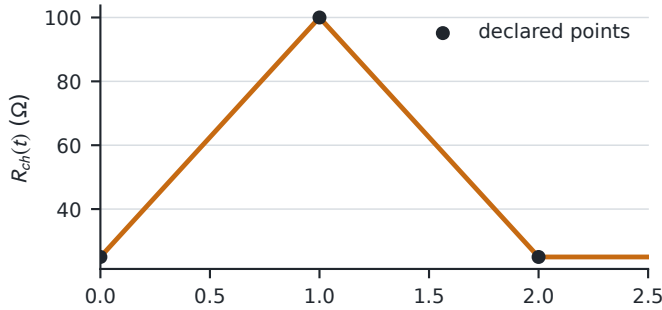
A prescribed component trajectory is used during the transient solve

The independent divider equation is evaluated at every saved time point; no fitted surrogate is used.

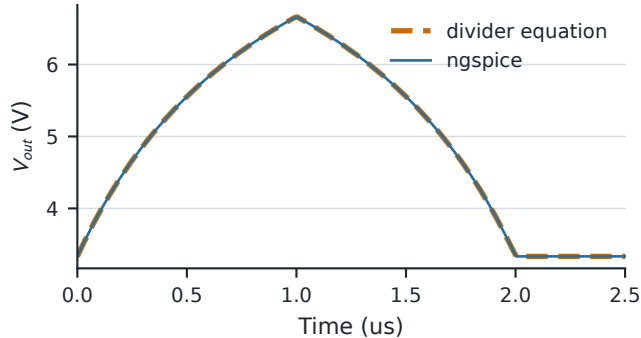
(a) Variable-resistor circuit



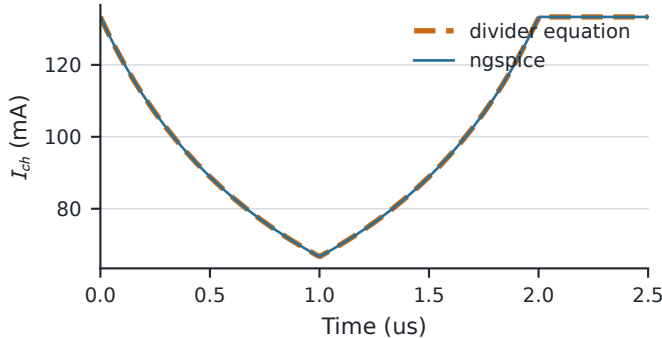
(b) Prescribed $R_{ch}(t)$



(c) Output voltage



(d) Chamber-element current



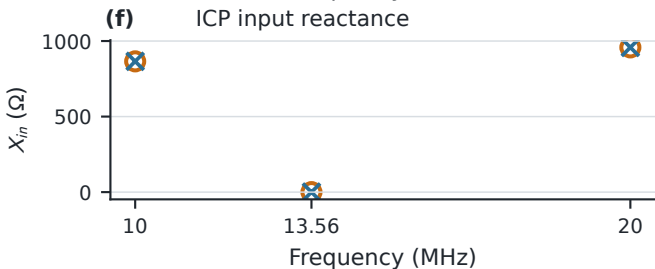
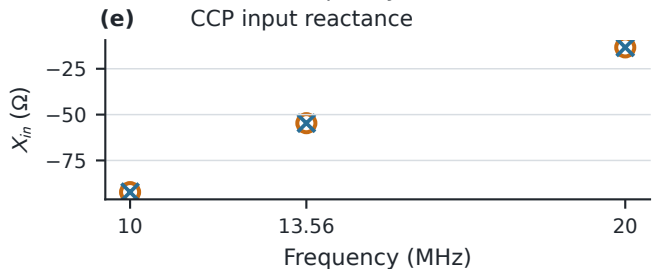
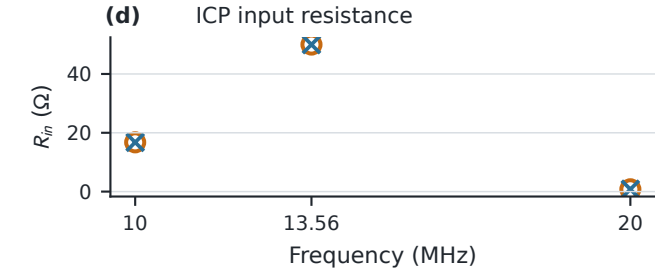
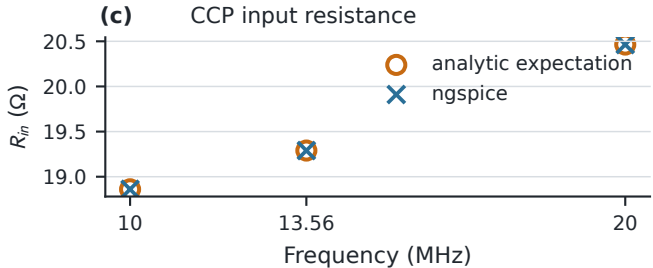
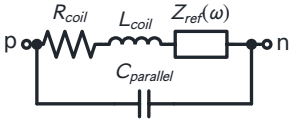
Effective plasma-load circuits propagate correctly through ngspice

Each declared load feeds the fixed benchmark L-match; points compare whole-network input Z, not plasma-state physics.

(a) Effective CCP terminal model



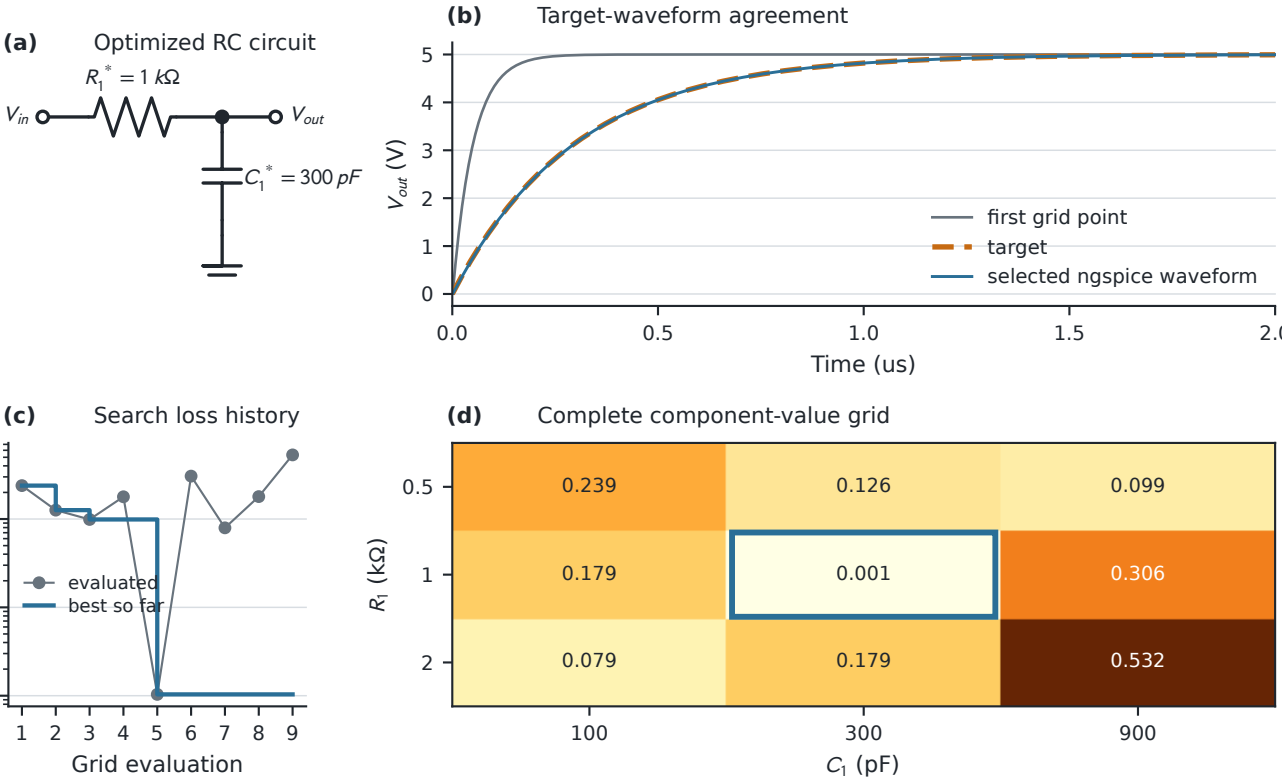
(b) Reduced ICP terminal model



Open circles: independent load + L-match equations. Crosses: ngspice extraction. Max |Z-component error|: CCP 4.58e-13 ohm; ICP 4.76e-10 ohm.

Element optimization recovers the RC time constant behind the target waveform

A finite 3 x 3 component grid is solved exhaustively; the selected waveform is an ngspice result, not a surrogate prediction.



Selected R1=1 kOhm, C1=300 pF, tau=300 ns; nRMSE=0.0010; 9/9 ngspice evaluations succeeded.